

In this project, AI (Gemini 3.1) was used to supplement the descriptions of game scenes, generate a small amount of dialogue, help me familiarize myself with Twine Harlowe 3.3.9, and create an interactive game map.

The last point is where AI made the greatest contribution to this project. Shigou's initial movement system was based on adding the option to move along with predetermined travel time to other passes at each location, and this connection was entirely based on my personal perception of traversability. Therefore, this was highly unreliable and time-consuming in the early stages of the game when there were over 30 locations passages. I confirmed with the AI whether it was possible to place interactive teleportation points and calculate the time using pixels on the image as coordinates. After receiving confirmation from the AI, I clarified the division of labor: I would create the map, design all locations, and record their pixel coordinates, while the AI would add the corresponding interactive points and location names to the game.

The map was created by taking 21 satellite images of Yubari in Mapbox, stitching them together using Photoshop, and drawing locations using GIMP and recording coordinates (blank map and map with points as well as the .xcf files f=can be found in /files).

Coordinates used in Mapbox:

43.007439/142.010257  
42.996149/142.010257  
42.984857/142.010257  
42.973563/142.010257  
42.962266/142.010257  
42.950968/142.010257  
42.939667/142.010257

43.007439/142.025695  
42.996149/142.025695  
42.984857/142.025695  
42.973563/142.025695  
42.962266/142.025695  
42.950968/142.025695  
42.939667/142.025695

43.007439/142.041132  
42.996149/142.041132  
42.984857/142.041132  
42.973563/142.041132  
42.962266/142.041132  
42.950968/142.041132  
42.939667/142.041132

List of all 42 locations and their coordinates, all locations are based on the Yubari.:

City Hall (1391, 5622)  
Shigou Clinic (1254, 5551)  
Apartment Entrance (1060, 5647)  
West River Arcade (1557, 5686)  
Kindergarten (1524, 5930)  
Shigou Primary and Middle School (1256, 5939)  
South Park (762, 5981)  
High School (565, 5551)  
Temple (862, 5175)  
Employment Center (1298, 4972)  
West River Mall (1515, 6340)  
Old Middle School (2118, 4704)  
Church (1385, 6475)  
East River Hot Spring (2385, 5341)  
Central Hot Spring (5848, 8524)  
Upper District Mall (1666, 4187)  
Pharmaceutical Company (1620, 7979)  
Upper District Arcade (1787, 3609)  
Fire Department (1575, 3713)  
Ski School (1878, 3159)  
Civic Center (1591, 3321)  
Police Station (1454, 2503)  
Lower District Mall (4887, 12778)  
West Lake Industrial Park (6673, 898)  
West River Industrial Park (1724, 6897)  
Power Station (2574, 2862)  
Central Arcade (3717, 9619)  
Lower District Arcade (4993, 12946)  
North Park (1788, 2780)  
Food Factory (1486, 4859)  
Post Office (4177, 9738)  
Lower District Clinic (5052, 12856)  
North Fields (3052, 6089)  
West Fields (1802, 8238)  
East Fields (5696, 9311)  
South Fields (4216, 10881)  
Coal Museum (120, 417)  
Observation Deck (4401, 7298)  
Old Upper District Station (1400, 3148)  
Old West River Station (1338, 5182)  
Old Central Station (3853, 9171)  
Lower District Station (4750, 12826)

|                                                                                                               |                                                                                                 |                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>UI_Time</b><br>{{**\$day   \$hour:<br>(if: \$minute <<br>10)[0\$minute]<br>(else:)]\$minute]  <br>Stamina: | <b>System_Time_Log</b><br>{ (set:<br>_stamina_loss to<br>(floor:<br>\$time_cost / 10))<br>(set: | <b>System_Calc_Time</b><br>{ (set: \$dx to<br>\$target_x -<br>\$current_x) (set:<br>\$dy to \$target_y<br>- \$current_y) |
| <b>Event Forced<br/>Sleep</b><br>{ (if: \$hour < 20<br>and<br>\$stamina_current<br>> 0) [(set: \$hour         | <b>Map UI</b><br>{ <span<br>id="map-<br>coords"<br>style="display:<br>none;"> <span             |                                                                                                                          |

Time systems, interactive map, and Stamina systems were all proposed by humans and implemented by AI. Time system is the first to be implemented, then interactive map. Stamina came much later in the development as I felt I need another restriction on exploration.

|                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| <b>System_Count_Sci</b><br>{ (set: _count to<br>0) (if:<br>\$has_makiri is<br>true)[(set:<br>_count to _count |
|---------------------------------------------------------------------------------------------------------------|

All the numerical settings were done by humans; AI was not involved at all. All scraps, their buffs, and the minimum requirement to trigger the ending are all made by human. AI is only involved in completing the descriptions of the scraps (details are in README). The Stamina recovery from sleep was initially fixed at 50. I later reduced that to  $25 + 0.1 * (\text{Lost Stamina})$  because I think this is cooler. Stamina reduction from movement is -1 Stamina every 10 minutes of movement, always rounded down. This is also the source of the 10-minute rest period for 1 Stamina. Players can only eat once every 4 hours because the Stamina recovery is twice that of rest. Players can only use the hot spring once every 6 hours because, in addition to restoring Stamina, it also provides a 10% movement speed boost for 2 hours.

```

{
  (set: $dx to $target_x - $current_x)
  (set: $dy to $target_y - $current_y)
  (set: $dist_straight to 0.5 * (sqrt: ($dx * $dx) + ($dy * $dy)))

  (set: $detour_k to 1.35 + (300 / ($dist_straight + 150)))
  (set: $dist_actual to $dist_straight * $detour_k)

  (set: _actual_speed to $move_speed_base * (1 + $speed_modifier))
  (set: $time_raw to $dist_actual / _actual_speed)
  (set: $time_cost to (round: $time_raw))

  (if: $time_cost < 1 and $dist_actual > 0)[
    (set: $time_cost to 1)
  ]
}

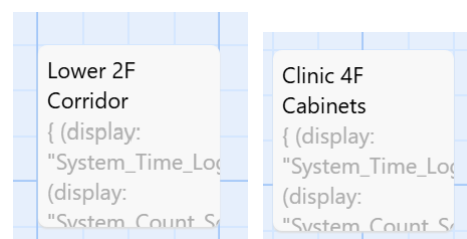
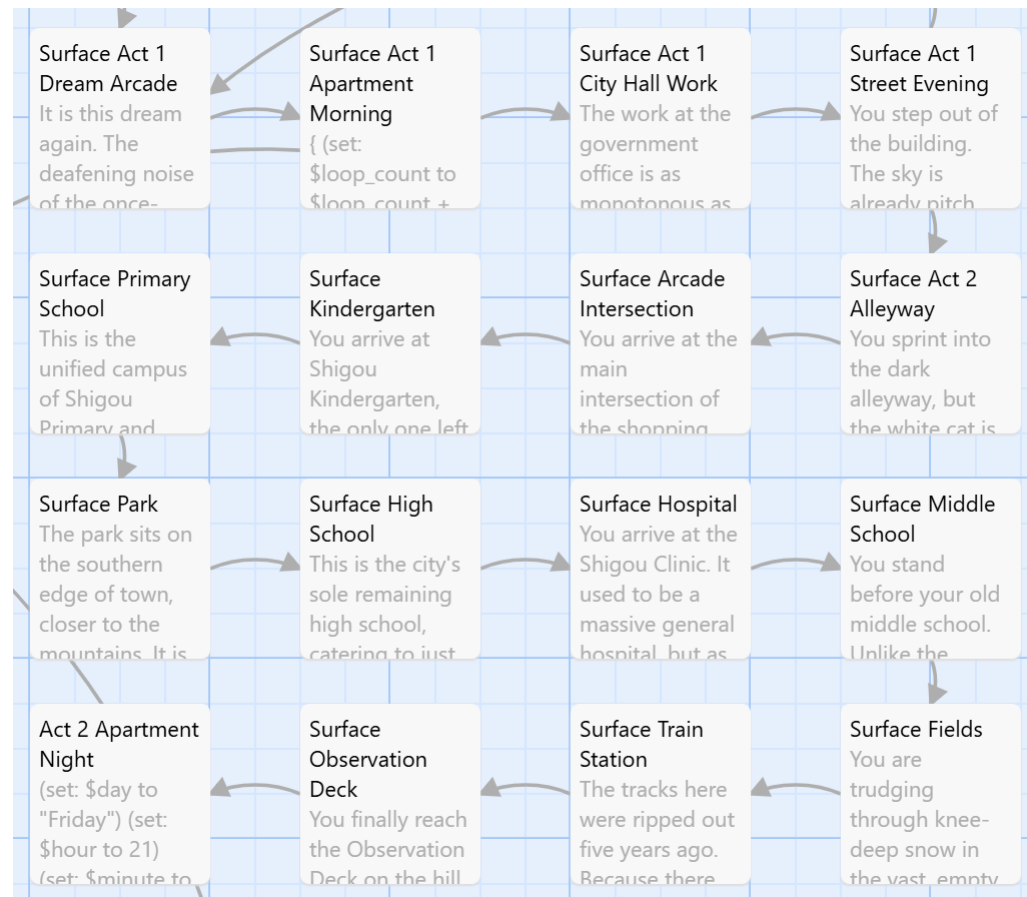
(set: $move_speed_base to 40)
(set: $speed_modifier to 0)
(set: $stamina_max_base to 100)
(set: $stamina_max to 100)
(set: $stamina_current to 100)
(set: $eat_cooldown_hour to 0)
(set: $bath_cooldown_hour to 0)
(set: $warm_expire_hour to 0)

```

The initial velocity was initially set to 46.67 mps (corresponding to 2.5 kph), but I felt it was too fast in actual use (as the game is set in the winter), so I changed it to 40 mps. A non-linearity coefficient (initially 1.35) was added because the Pythagorean theorem can only calculate straight-line distances, and the inverse relationship between the non-linearity coefficient and distance aligns with reality. Notice that a non-linearity coefficient of 1.35 is not high, because most cities in Hokkaido were established during the late 19th and early 20th centuries, and therefore, the initial city plans deliberately created a grid-like structure.

The location setting and exploration are entirely designed by humans, including how scraps are obtained (see several examples in the README for details). The events triggered by probability are also entirely set by humans. For example, the scraps and figures in the fields are entirely set by humans: this is the exploration with the greatest randomness in the game. At game initialization, one of the four fields is selected as the location to obtain the scrap and figure; the probability of triggering exploration in the field is 40%; the steps required to reach the destination are also completely random, and failure twice results in exploration failure. The initial design concept was to create a terrifying atmosphere and guide the player through the volume of laughter audio, but it was found during testing that the audio needed to be played manually by the player, so this approach was abandoned and text prompts were used instead. The audio was not added to the game, which is a pity. Another example is the Observation Deck's search for the amusement park: this was inspired by a video I saw about searching for the lost amusement park in Yubari, but the exploration method here is fixed (3 times east, 1 time north) because I initially thought the amusement park should be in a fixed

location, but if I were to set it now, the exploration here would also become random. The exploration here costs 2 Stamina because it is in a forest in the snow.



|                                                                                                  |                                                                                             |                                                                                                |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Old Upper District Station</b><br>{ (set: \$current_x to 1400, \$current_y to 3148) (display: | <b>Upper Station Old Man</b><br>{ (set: \$minute to \$minute + 10) (set: \$hour to \$hour + | <b>Upper Station Ramen Exec</b><br>{ (set: \$minute to \$minute + 20) (set: \$hour to \$hour + |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|

|                                                                                                           |                                                                                     |                                                                                        |                                                                                      |                                                                                       |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Field Event Init</b><br>{ (set: \$minute to \$minute + 10) (set: \$hour to \$hour + (floor: \$minute / | <b>Field Event Nav</b><br>{ (display: "System_Count_Sc (set: _current_dist to (abs: | <b>Field Event Move</b><br>{ (set: \$minute to \$minute + 10) (set: \$hour to \$hour + | <b>Field Event Fail</b><br>(display: "UI_Time") --- You wander through the deep snow | <b>Field House Outside</b><br>(display: "UI_Time") --- Following the voice you arrive |
| <b>Field House Leave Story</b><br>{ (set: \$has_girl_figure to true) } (display:                          |                                                                                     |                                                                                        |                                                                                      |                                                                                       |

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|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Museum Mine Registration</b><br>{ (set: _error_msg to "") } You follow the old man   | <b>Museum Mine Gates</b><br>{ (set: \$met_girl_in_mine to true) } The old man looks                 | <b>Mine Path 1</b><br>{ (set: \$minute to \$minute + 10) (set: \$hour to \$hour + (floor: \$minute / | <b>Mine Path 2</b><br>{ (set: \$minute to \$minute + 5) (set: \$hour to \$hour + (floor: \$minute / 60) | <b>Mine Path 3</b><br>(display: "UI_Time") --- You turn left, and the beam of your headlamp             | <b>Girl Dialogue 1</b><br>"I was created right here in Shigou exactly 75 years ago by a mysterious | <b>Girl Dialogue 2</b><br>"My abilities operate under very strict boundaries," she explains, looking |
| <b>Girl Dialogue 3</b><br>"Why didn't I just give you the memories directly? Because my | <b>Girl Dialogue 4</b><br>"Many young outsiders come here wishing to explore Shigou's history," she | <b>Girl Dialogue 5</b><br>"A young person like you... you seem to genuinely love the                 | <b>Girl Dialogue Gift Exec</b><br>{ (set: \$has_lotus_bloom to true) (display: "System_Count_Sc         | <b>Mine Exit Exec</b><br>{ (set: \$minute to \$minute + 35) (set: \$hour to \$hour + (floor: \$minute / | <b>Museum Old Man Chat</b><br>{ (set: \$minute to \$minute + 10) (set: \$hour to \$hour +          | <b>Mine Special Scrap Exec</b><br>{ (set: _show_scrap to false) (if: \$slaver_name.is_               |

All plot-related descriptions and long dialogues were written entirely in Chinese by humans, then translated and polished by AI, and finally reviewed and revised by humans. Something about Old Upper District Station not mentioned in README:

Out on the snow-covered platform, next to the frozen tracks, stands a single weathered sign that reads: *\*"Thank you to the Shigou railway."\**

This also came from real old Yubari Station, where it states "Thank you to the Yubari railway," but, of course, is written in Japanese.

This is the unified campus of Shigou Primary and Middle School. As the population naturally plummeted to barely 6,000, the pragmatic city government consolidated both schools into this single, older primary school building to save on heating and maintenance. It currently houses 126 primary students and 154 middle schoolers. The weekend lights are already off.

This is the city's sole remaining high school, catering to just 64 students. You remember walking these very halls, fully aware that most of your classmates would inevitably leave Shigou for larger cities. That high outmigration rate is just a simple fact of life here, neither good nor bad.

### **\*\*History Exhibition: The Rise and Fall of the Population\*\***

Moving to the next corridor, striking line graphs cover the walls, illustrating the dramatic rise and devastating fall of Shigou's population. The data on occupational distribution paints a vivid, brutal picture of the city's economic shift:

\* **\*\*1960:\*\*** The absolute peak. **\*\*48.4%\*\*** of the population was employed directly in the mining industry, while only 11.5% worked in the service sector. The city was a roaring engine of black gold.

\* **\*\*1985:\*\*** The painful decline. The mining population dropped to **\*\*26.3%\*\***, and the service sector grew to 19.2% as the government desperately tried to pivot the local economy.

\* **\*\*2005:\*\*** The final collapse. A staggering **\*\*0.1%\*\*** remained in mining—mostly for structural maintenance of abandoned pits—while 38.5% clung to a service sector.

These data were entirely set by humans and are based on real-world data from Yubari.

The protagonist was initially designed as a character without a displayed gender or name, because I thought that was cool and adds ambiguity. The protagonist was initially conceived as a city hall employee, but I later learned that employees in a declining city couldn't possibly leave at 5 PM, and leaving too late would prevent players from exploring during weekdays. Therefore, I changed it to working at the front reception desk, a job where leaving at 5 PM is highly likely, and this overtime setting was incorporated into the background description.

The game's reasoning for forcing players to return home at 8 PM is that there are no streetlights at night. However, in reality, such cities don't have streetlights at all at night, and it gets dark early in winter. Exploration should actually be prohibited at 7 PM or even earlier, but that would make weekday exploration almost unplayable, so it was changed to 8 PM.

As stated in the README, many of the game's settings are inspired by Susumu Hirasawa's song "Byakkoya no Musume." This is quite obvious if you've seen the lyrics. The AI has almost no impact on the world-building, plot, or setting; its role is limited to helping me look up information from reality.